THEXRA ENTERPRISE SOLUTION PROPOSAL

Architecture Recommendation Brief

Date: 01/10/2026
Status: Draft

1. CLIENT PROBLEM STATEMENT

Surgical trainees lack sufficient hands-on procedural practice and technical mastery due to restricted physical access and high operating costs of specialized wet labs.

2. AI RATIONALE & ANALYSIS

"A combination spatial solution circumvents physical lab bottlenecks by providing immersive, repeatable surgical simulations with automated technical evaluation. THEXRA Suite gives clinical educators an integrated ecosystem to author realistic anatomical scenarios, deploy simulations to headsets, and remotely evaluate trainee performance."

3. PROPOSED SOLUTION CONCEPT

An enterprise spatial surgical training platform combining photorealistic XR procedure simulations, automated AI skill evaluation, and real-time remote mentor proctoring.

IMPLEMENTATION APPROACH:

1. Phase 1: Curriculum alignment, procedural mapping, and anatomical 3D asset modeling.
2. Phase 2: Core simulation build, spatial tool interaction physics, and automated scoring integration.
3. Phase 3: Clinical cohort pilot testing, system validation, and institutional deployment.

DETAILED IMPLEMENTATION PLAN:

1. **Discovery** (2 weeks):Curriculum requirement document • Technical architecture blueprint
2. **Design** (2.5 weeks):Surgical workflow wireframes • 3D anatomical visual specification
3. **Prototype** (3 weeks):Interactive simulation prototype • Early trainee feedback report
4. **Development** (5 weeks):Functional surgical simulation modules • Integrated AI scoring backend
5. **Testing** (2 weeks):Clinical accuracy review sign-off • System performance and load test log
6. **Deployment** (1.5 weeks):Production system installation • Trainer and administrator user guide
7. **Support** (Ongoing):Continuous platform maintenance • Quarterly trainee progress analytics report

4. TECHNOLOGY ARCHITECTURE BLUEPRINT

Combination solution — THEXRA Suite (Remote, Make, View)

AUTOMATED SYSTEM FLOW DIAGRAM:



1. Overview: The solution integrates spatial XR headsets with an AI-driven simulation runtime to replicate operative procedures in a risk-free virtual setting. Trainee actions and instrument trajectories are captured in real time and synced to a centralized cloud portal for instructional scoring.

2. User Journey: Trainee launches the XR application and selects a procedural module → Follows interactive anatomical guidance and handles simulated instruments → AI tracks precision and technique →

> Trainee completes the surgery and views an instant debrief → Supervisor reviews scoring on the dashboard.
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> **3. Hardware:** HTC VIVE Eagle XR Headset, Precision Haptic Motion Controllers
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> **4. Software:** THEXRA View XR Simulation Client, THEXRA Make Scenario Authoring Tool, THEXRA Remote Tele-Mentoring Module
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> **5. AI Components:** Spatial Instrument Tracking and Accuracy Scorer, Procedural Sequence and Error Detection Model
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> **6. XR Components:** Photorealistic 3D Anatomical Models, Physics-Based Surgical Interaction Engine
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> **7. Backend:** Cloud Simulation Telemetry Service, Secure Role-Based Access and LMS Integration API
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> **8. Dashboard:** Instructor Procedure Review and Scoring Portal, Trainee Competency Matrix and Analytics Console
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> **9. Data Flow:** XR Headset Sensors → Local Physics Runtime → Cloud Telemetry API → AI Performance Engine → Web Dashboard
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> **Alternative Option:** MR — Microsoft HoloLens 2

5. EXPECTED BUSINESS OUTCOMES

Enables unlimited repeatable surgical practice sessions, lowers physical lab operating costs by up to 50%, and automates procedural compliance tracking.

6. IMPLEMENTATION NEXT STEPS

Conduct a scoping workshop with clinical faculty to map priority surgical workflows and curriculum milestones.